IndicDoc Verify: Layout-Aware Pre-Print Verification of Multilingual Indian Question Papers

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***Abstract*—Indian examinations are set in one language and translated into Hindi and regional languages, and every version is checked by hand; errors keep reaching candidates, from 49 wrong questions in the Tamil version of NEET 2018 to re-tests of three UGC-NET papers in 2026. We present IndicDoc Verify, an offline system that reads a printed question paper in any of 12 Indian languages and flags structural and numerical errors, and every difference from its source version, before printing. A deep-learning layout detector recovers questions, options and answers (AP@50 0.81–0.90); Tesseract reads the text with an English number pass that must agree on every digit; questions are built using the layout, checked for numbering, duplicates, options, marks and repeats, and two language versions are aligned and compared without translation. On 132 English–Hindi and English–Tamil paper pairs with injected errors, read with real OCR, the system finds 98% of errors at the right question on born-digital papers (98% on degraded scans) with 3.0 false alarms per clean pair; the number pass alone cuts false alarms from 13.8 to 4.5.**

***Index Terms*—document layout analysis, Indian languages, OCR, examinations, translation quality, verification.**

I. Introduction

National and state examinations in India are multilingual: NEET-UG is offered in 13 languages, and the Union and state Public Service Commissions, school boards and universities print bilingual papers. A paper is set in one language, translated, and every version is vetted by hand. When that step fails, candidates who chose a regional language sit a different paper. In NEET-UG 2018, 49 questions of the Tamil version were wrong and the Madras High Court ordered 196 grace marks, later set aside by the Supreme Court [15]; the Karnataka civil-services preliminary paper of 2024 had 59 errors in Kannada and was re-held, with errors again [16]; three UGC-NET papers of June 2026 were re-tested after translation errors and questions repeated from earlier papers [17]; an MBBS paper of 2026 printed the same MCQ twice [18].

We found no tool that checks a paper and its translations automatically. Translation-quality estimation [8] and LLM back-translation [7] work on plain sentences and cannot see question numbers, options or marks; the IndicDLP layout dataset [1] labels questions and options in 12 languages, but detected structure has not been used to verify a document; and a pre-print paper is too confidential for a cloud service. (Court-filing scrutiny, by contrast, already uses AI defect detection [19].)

Our contributions: (1) IndicDoc Verify, an offline pipeline that turns a printed paper into structured questions with a deep-learning layout detector and checks it; (2) *translation parity without translation*: language versions are aligned and compared on numbers, options, marks, sub-questions and figures, which are the same in every language; (3) a *number pass* in which two OCR models must agree on every digit; (4) a benchmark of 132 rendered paper pairs with nine kinds of real error, a separate development set, and ablations.

II. Related Work

Indian-language documents. IndicDLP [1] provides 119,806 pages with 42 labels and benchmarks modern detectors; DocLayout-YOLO [11] adapts YOLOv10 to documents; RoDLA [10] studies layout robustness. Recognition work covers handwritten Indic text [2], Hindi OCR with vision-language models [6] and tables [3]; multimodal models [4], [5] need reliable OCR. MILU [9] collects Indian exam questions in 11 languages for evaluating LLMs.

Checking translations. Quality estimation such as CometKiwi [8] scores a translation without a reference, sentence by sentence. Babayeva *et al.* [7] translate a physics test with an LLM and still review each item by hand. Neither uses the page structure of an exam paper, which is where many costly errors lie (a missing option, a changed number, wrong marks).

III. IndicDoc Verify

A. Layout. Each page (PDF rendered at 200 dpi) goes through DocLayout-YOLO [11], fine-tuned on IndicDLP by its authors, at a high-recall threshold chosen on validation pages. We also fine-tuned YOLOv8s [12] and Faster R-CNN [13] on a 2,293-page language-balanced subset to choose the detector (Table I). The released model recovers the question-paper labels well (Table II).

TABLE I
LAYOUT DETECTION ON 682 INDICDLP TEST PAGES

Detector	mAP@50	mAP@50:95	F1	s/page
Faster R-CNN (ours)	0.148	0.079	0.469	0.202
YOLOv8s (ours)	0.259	0.165	0.551	0.024
DocLayout-YOLO [11]	0.759	0.575	0.810	0.150

TABLE II
AP@50 ON QUESTION-PAPER LABELS

Label	DocLayout	YOLOv8s	F. R-CNN
first-level q.	0.85	0.42	0.24
second-level q.	0.90	0.04	0.02
third-level q.	0.87	0.03	0.00
options	0.84	0.55	0.24
answer	0.81	0.00	0.00

B. OCR with a number pass. Tesseract [14] reads the page with the model of its own script. Indic models misread Latin digits in mixed text ("1947" as "947", "1" as the danda "।", "A" as "८"). An English pass re-reads the page; each numeric token replaces the overlapping word of the Indic reading when that word has no Indic letters or the English reading is more confident, is inserted where the Indic reading has nothing, and a Latin number of the Indic reading that the English pass does not confirm is dropped. Pages printed with Indian digits are detected and keep them.

C. Questions. A first-level-question region starts a question (or a text region beginning with a question number, if the detector missed it); sub-questions, options, figures, tables and formulas follow it. Options are split by the horizontal gaps between words inside an option region, so a garbled marker does not change the count; marks ("[2 marks]", "[2 अंक]", "(2 மதிப்பெண்கள்)") and the declared maximum are read in all 12 languages.

D. Checks. On one paper: numbering gaps and repeats, duplicated questions (character 3-gram cosine ≥ 0.9 with identical numbers), option count, repeated or empty options, marks against the declared maximum, page numbering. Against a question bank: repeats (cosine ≥ 0.8, identical numbers). Between versions s and t, questions are aligned by Needleman–Wunsch with the score

$$\sigma(i, j) = 2\,J(N_i, N_j) + \nu(i, j) + 0.5\cdot[k_i = k_j], \quad \text{gap} = -0.6$$

where N is the multiset of numbers in a question and its options, J their Jaccard similarity, $\nu = 1\ (-0.5)$ if the printed numbers agree (differ), and k the skeleton (option count, sub-questions, marks, visuals). Aligned pairs are compared on every number in the stem and options, option count, marks, sub-questions and visuals; unaligned questions are reported missing or extra. A cross-lingual similarity model can be plugged in as an extra meaning check. Every finding carries its question, page and box.

E. Format profiles and deployment. The same engine checks any document against a JSON profile of required elements, order and bilingual text (e.g. an Office Memorandum). The system runs offline in one Docker image with a web app and a REST API; the report outlines each finding on its page (Fig. 1).

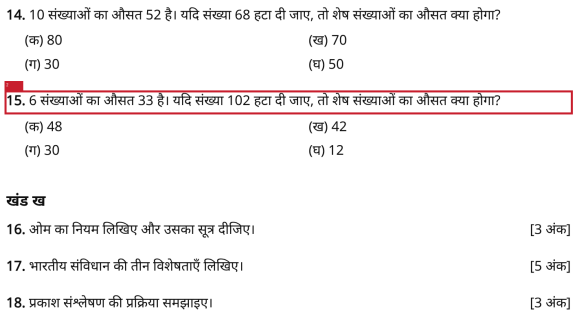

14. 10 संख्याओं का औसत 52 है। यदि संख्या 68 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?
(क) 80 (ख) 70
(ग) 30 (घ) 50
15. 6 संख्याओं का औसत 33 है। यदि संख्या 102 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?
(क) 48 (ख) 42
(ग) 30 (घ) 12
खंड ख
16. ओम का नियम लिखिए और उसका सूत्र दीजिए। [3 अंक]
17. भारतीय संविधान की तीन विशेषताएँ लिखिए। [5 अंक]
18. प्रकाश संश्लेषण की प्रक्रिया समझाइए। [3 अंक]

Fig. 1. A changed number in the Hindi version of a benchmark paper, found and outlined on the page by IndicDoc Verify.

IV. BENCHMARK

Erroneous pre-print papers are not public, so we built a benchmark. Each pair is an English source and its Hindi or Tamil version: 15 MCQs (numeric templates and fixed questions) and 3 short-answer questions with marks, rendered at 200 dpi with Noto fonts by a browser, which also records each region's box. 108 of the 132 test pairs carry one injected error (Table III); 24 are clean. Half are degraded like scans (blur 0.7 px, Gaussian noise $\sigma = 8$, JPEG 55). Pages are read with Tesseract 5 (tesseract.js, best-integer models). Layout boxes come from the renderer (an oracle layout); a *noisy-layout* run shifts boxes by up to 8 px and relabels 10% of question and option regions as paragraphs. An error counts as found only if a finding of the matching kind points at the right question (± 1) or, for paper-level errors, at the paper. The parser was developed on a separate 48-pair development set; the test set was run once.

TABLE III
ERROR TYPES INJECTED IN THE BENCHMARK

Error	Pairs	Found
Number changed in stem	12	100%
Number changed in an option	12	100%
Option missing	12	100%
Question missing	12	100%
Question printed twice	12	100%
Marks differ	12	100%
Wrong maximum marks	12	83%
Question numbering error	12	100%
Repeated from a past paper	12	100%

V. RESULTS

Errors found and false alarms. Table IV: on born-digital papers, the normal pre-print case, IndicDoc Verify finds 98% of injected errors at the right question with 3.00 false alarms per clean pair; scan degradation lowers this to 98%

with 5.50 false alarms. Fig. 2 shows the result per error type.

TABLE IV
VERIFICATION RESULTS (TEST SET)

Condition	Found	FA / clean pair
Born-digital	98%	3.00
Scan-degraded	98%	5.50
English → Hindi	96%	7.00
English → Tamil	100%	1.92
All	98%	4.46

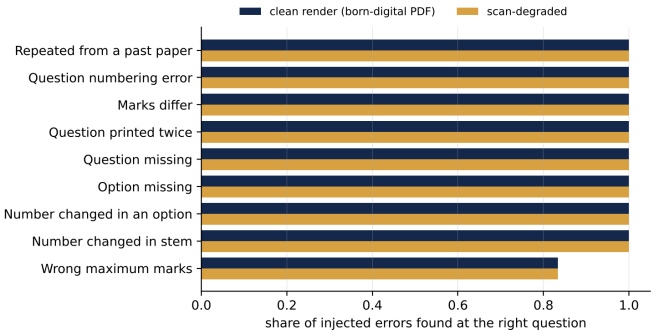


Fig. 2. Injected errors found at the right question, per error type, born-digital vs. scan-degraded pages.

Ablations. Table V and Fig. 3: removing the number pass raises false alarms to 13.75 per clean pair; splitting options by their markers instead of the layout raises them to 11.46. A noisy layout still finds 98% of errors, because questions are recovered from their text when a label is wrong.

TABLE V
ABLATIONS (TEST SET)

System	Found	FA / clean pair
Full system	98%	4.46
No number pass	77%	13.75
No layout-based options	98%	11.46
Noisy layout	98%	7.50

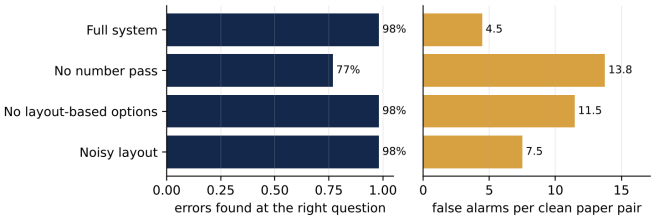


Fig. 3. Ablations: errors found (left) and false alarms per clean pair (right).

Error analysis. Most remaining false alarms are option values that OCR dropped or merged on one side ("216" read as "26"), more often on degraded Hindi pages. The only two misses were Hindi papers whose maximum-marks line OCR read as "पूर्णांक" (anusvara dropped). After the test run we widened that pattern; re-running the unchanged test set then finds 100% of errors with 4.38 false alarms per clean pair (post-hoc; the numbers above are from the frozen system).

VI. Discussion and Limitations

The checks need no translation, so they work for any pair of the 12 languages and cannot be fooled by fluent but wrong text in the way a quality-estimation score can. Their blind spot is a change of meaning that keeps numbers and structure (a dropped word, as in NEET 2021); the meaning hook needs a cross-lingual model that is evaluated on exam language. The benchmark uses rendered papers and oracle layout boxes; the layout model's own errors are covered by the noisy-layout run and by its measured AP, and a pilot on real pre-print papers of an exam body is the next step. Number words and formulas are not compared yet.

VII. Conclusion

IndicDoc Verify brings deep-learning layout analysis to a problem that is costly, recurring and so far unautomated: checking an Indian question paper and its translations before printing. It finds most injected errors at the right question with few false alarms on born-digital papers, runs offline, and covers 12 languages. Code, benchmark generator and results are open.

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